

A-level

# ENVIRONMENTAL SCIENCE

## Questions by topic: Fluctuations in supply/demand & energy storage - Set 1

### Mark scheme

AQA spec 3.3.4.1; Energy resources; textbook pp. 217-224

### Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 01.1     | <ul style="list-style-type: none"> <li>• Lowest overnight (about 25 GW at 04:00), rising steeply from 06:00 to about 40 GW by 08:00</li> <li>• Fairly steady through the day then a peak of about 50 GW at 18:00 (evening) before falling overnight</li> </ul>                                                                                                                                                                                                                         | 2     |
| 01.2     | <ul style="list-style-type: none"> <li>• 24-hour day / night cycle: people asleep overnight; morning rise as heating, lighting, cooking, industry and offices start</li> <li>• Evening peak: lighting, cooking, heating as people return home; short-term 'TV pickup' at programme breaks / mealtimes</li> <li>• Other causes: seasonal (winter heating / lighting, summer air conditioning); weather-related (cold snaps); weekday / weekend differences in industrial use</li> </ul> | 4     |
| 01.3     | <ul style="list-style-type: none"> <li>• They generate 24 hours a day because it is uneconomic / slow to turn them off and on, and output changes slowly</li> <li>• Demand falls faster at night than output can be reduced, so the surplus must be stored or is lost as heat from cables</li> </ul>                                                                                                                                                                                   | 2     |
| 01.4     | <p><i>Any 2 from:</i></p> <ul style="list-style-type: none"> <li>• Intermittent renewable resources: wind, solar, tidal</li> <li>• Bulk delivery of transported fuels (oil tankers, coal trains, biofuel harvests)</li> <li>• Breakdowns / maintenance of power stations</li> </ul>                                                                                                                                                                                                    | 2     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 221-222; AQA spec 3.3.4.1.</p>                                                                                                                                                                                                                                                                                                                                                     |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 02.1     | <ul style="list-style-type: none"> <li>• At times of low demand / surplus supply (night, windy periods) surplus electricity pumps water from the lower to the upper reservoir, storing it as gravitational potential energy</li> <li>• At peaks in demand water flows down through the turbines generating electricity</li> <li>• Can go from standby to full power in under 15 seconds so it meets sudden peaks (TV pickup) that fossil / nuclear stations cannot follow</li> </ul>                                    | 3     |
| 02.2     | <ul style="list-style-type: none"> <li>• Mass = <math>7 \times 10^6 \times 1000 = 7 \times 10^9</math> kg</li> <li>• PE = <math>m g h = 7 \times 10^9 \times 9.8 \times 500</math></li> <li>• = <math>3.43 \times 10^{13}</math> J (accept <math>3.4 \times 10^{13}</math>)</li> </ul>                                                                                                                                                                                                                                  | 3     |
| 02.3     | <ul style="list-style-type: none"> <li>• The electricity used for pumping is surplus that would otherwise be wasted / lost as heat, or cheap off-peak electricity</li> <li>• It avoids building rapid-response low-efficiency peaking plants (open-cycle gas turbines) and allows intermittent renewables to be used</li> </ul>                                                                                                                                                                                         | 2     |
| 02.4     | <p>Any 2 from:</p> <ul style="list-style-type: none"> <li>• Two water bodies / sites with a large height difference close together (mountainous)</li> <li>• Impermeable, stable rock</li> <li>• Close to the grid / demand</li> <li>• Adequate water supply</li> </ul>                                                                                                                                                                                                                                                  | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 222; AQA spec 3.3.4.1.                                                                                                                                                                                                                                                                                                                                                                                                     |       |
| 03.1     | <ul style="list-style-type: none"> <li>• Surplus electricity (eg windy nights) electrolyses water to produce hydrogen which is stored</li> <li>• Hydrogen has a high energy density so it can be transported / stored for long periods</li> <li>• Released by combustion for heat, industrial processes or steam turbines; or by fuel cells combining hydrogen and oxygen electrochemically to produce electricity and water</li> <li>• Can also be used as a vehicle fuel or converted to methane / ammonia</li> </ul> | 4     |
| 03.2     | <p>Any 3 from:</p> <ul style="list-style-type: none"> <li>• Compressed gas (up to 700 times normal volume) - energy to run the compressor, strong tanks</li> <li>• Liquefied at very low temperature - energy for refrigeration and compression</li> <li>• Adsorbed on a metal matrix (metal hydride) at low pressure - tanks larger and heavier than a petrol tank storing the same energy</li> <li>• Conversion to ammonia - some energy needed but less than the alternatives</li> </ul>                             | 3     |
| 03.3     | <ul style="list-style-type: none"> <li>• Both convert chemical energy to electricity electrochemically, but a battery stores a fixed amount of chemicals which are reformed by recharging</li> <li>• A fuel cell keeps producing electricity as long as fresh fuel (hydrogen / alcohol) and oxidant are supplied</li> </ul>                                                                                                                                                                                             | 2     |
| 03.4     | <ul style="list-style-type: none"> <li>• Using hydrogen made from surplus renewable electricity as the main store and carrier of energy, released when demand exceeds supply or intermittent sources fail, for heating, vehicles and electricity</li> </ul>                                                                                                                                                                                                                                                             | 1     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 219-221; AQA spec 3.3.4.1.                                                                                                                                                                                                                                                                                                                                                                                                 |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 04.1     | <ul style="list-style-type: none"> <li>• Vehicles: high energy density (<math>200 \text{ Wh kg}^{-1}</math>) minimises battery mass so range is greater; many cycles and high efficiency</li> <li>• Stationary storage: mass is unimportant so the low energy density of lead-acid does not matter; lowest cost per kWh; well-established, recyclable</li> <li>• Lead-acid drawbacks: fewer cycles and lower efficiency; lithium fire risk needs management</li> <li>• Nickel-cadmium largely replaced because of cadmium toxicity and cost</li> </ul> | 4     |
| 04.2     | <p>Any 2 from:</p> <ul style="list-style-type: none"> <li>• Recharging speed</li> <li>• Self-discharge during non-use</li> <li>• Availability of raw materials (lithium, cobalt)</li> <li>• Recyclability / disposal</li> </ul>                                                                                                                                                                                                                                                                                                                        | 2     |
| 04.3     | <ul style="list-style-type: none"> <li>• Parked electric vehicles plugged into the grid form a very large distributed battery</li> <li>• At peaks a small proportion of their stored energy is fed back to the grid; surplus grid energy recharges them - avoiding costly, low-efficiency open-cycle gas turbines</li> </ul>                                                                                                                                                                                                                           | 2     |
| 04.4     | <ul style="list-style-type: none"> <li>• Surplus electricity electrolyses water to hydrogen, which can be converted to methane</li> <li>• Methane is fed into the existing natural gas pipe network and storage, so no new distribution system is needed</li> </ul>                                                                                                                                                                                                                                                                                    | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 223-224; AQA spec 3.3.4.1.                                                                                                                                                                                                                                                                                                                                                                                                                                |       |
| 05.1     | <ul style="list-style-type: none"> <li>• Surplus energy drives a pump compressing air stored in underground caverns (salt mines); releasing it drives machinery / turbines</li> <li>• Compression produces a lot of heat; storing this heat (hot oil / molten salt) and returning it when the air expands recovers energy that would otherwise be lost</li> </ul>                                                                                                                                                                                      | 2     |
| 05.2     | <ul style="list-style-type: none"> <li>• As volume increases the surface area : volume ratio decreases</li> <li>• Heat is lost through the surface so there is less surface per unit of stored heat; very large stores can carry summer solar heat into winter (inter-seasonal storage)</li> </ul>                                                                                                                                                                                                                                                     | 2     |
| 05.3     | <ul style="list-style-type: none"> <li>• Potassium nitrate has a much higher boiling point so it stores heat at up to <math>550^\circ\text{C}</math> without pressurisation</li> <li>• Water could only reach such temperatures under high pressure, increasing costs; high temperature needed to raise steam for turbines</li> </ul>                                                                                                                                                                                                                  | 2     |
| 05.4     | <ul style="list-style-type: none"> <li>• Materials with high specific heat capacity (concrete, water) warm and cool slowly</li> <li>• They absorb heat in hot weather reducing overheating / air conditioning, and release it as the weather cools, delaying the need for heating</li> </ul>                                                                                                                                                                                                                                                           | 2     |
| 05.5     | <p>Any 2 from:</p> <ul style="list-style-type: none"> <li>• Flywheels: rotating mass stores kinetic energy for short-term smoothing / generation</li> <li>• Supercapacitors: electrochemical capacitors with 10-100 times the energy density of conventional capacitors; possible future large-scale electricity storage</li> <li>• Thermal storage in molten salt / high-volume tanks; hydrogen / P2G; batteries</li> </ul>                                                                                                                           | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 222-224; AQA spec 3.3.4.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |
| <b>06.1</b> | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Renewables are intermittent (solar at night, calm days, slack tides) and some unpredictable; demand fluctuates daily, weekly, seasonally and short-term; supply and demand rarely match</li> <li>• Without storage, surplus is wasted and shortages must be met by fossil-fuel peaking plants; large-scale electricity storage not yet possible directly so energy must be converted</li> <li>• Pumped-storage HEP: proven, rapid response, peak shaving, but limited sites</li> <li>• Batteries: lithium-ion for grid and vehicles; factors affecting viability; V2G</li> <li>• Hydrogen: electrolysis, storage methods, fuel cells, hydrogen economy, P2G into the gas grid; high energy density enables vehicle use</li> <li>• Compressed air with heat recovery; thermal storage (molten salt for CSP, high-volume stores, thermal mass); flywheels; supercapacitors</li> <li>• Grid management / ICT and diversity of resources (wind + solar + tidal + biofuel) reduce but do not remove the need for storage; energy conservation reduces peaks</li> <li>• Evaluation of efficiency losses, costs, resource limits (lithium), and which technologies suit short-term vs inter-seasonal storage; judgement</li> </ul> | 9     |
| <b>06.2</b> | <ul style="list-style-type: none"> <li>• Storing surplus energy at times of low demand / high supply to meet later peaks in demand, so peak generating capacity can be reduced</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 217-224; AQA spec 3.3.4.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |